

CAPSTONE PROJECT PROPOSAL FALL 2025



Adaptive CCTV Compression System Using Dynamic Region Of Interest (ROI) Control

Submitted by

Shakibul Islam Tamim

ID: 0822220105101046

Jannatul Tabassum Nahar

ID: 0822220205101001

Mostafa Main Uddin

ID: 0822220105101017

Supervised by

Hasan Abdullah

Assistant Professor, Dept. of CSE, BAIUST

in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND
TECHNOLOGY (BAIUST)**

DECLARATION

We hereby declare that the work presented in this Capstone Project Proposal is the result of our own investigation and efforts, conducted under the guidance and supervision of Hassan Abdullah, Assistant Professor, Department of Computer Science and Engineering, Bangladesh Army International University of Science and Technology, Cumilla, Bangladesh. We confirm that no part of this proposal has been submitted, nor is being submitted, for the award of any other degree or diploma at any institution.

Countersigned

Signature

(Hasan Abdullah)
Assistant Professor

Shakibul Islam Tamim
ID: 0822220105101046

Supervisor

Jannatul Tabassum Nahar
ID: 0822220205101001

Mostafa Main Uddin
ID: 0822220105101017

APPROVAL

The Capstone Project titled “Adaptive CCTV Compression System Using Dynamic Region Of Interest (ROI) Control,” submitted by Shakibul Islam (0822220105101046), Jannatul Tabassum Nahar (0822220205101001), and Mostafa Main Uddin (0822220105101017) to the Department of Computer Science and Engineering, Bangladesh Army International University of Science and Technology, has been reviewed and approved as satisfactory for partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

Board of Examiners

Hasan Abdullah

Assistant Professor

Department of Computer Science and Engineering, BAIUST

Supervisor

(Member Name and Signature)

Department of Computer Science and Engineering, BAIUST

Member 1

(Member Name and Signature)

Department of Computer Science and Engineering, BAIUST

Member 2

(Member Name and Signature)

Department of Computer Science and Engineering, BAIUST

Member 3

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1. Introduction

Surveillance systems are essential for security in smart cities, institutions, and public spaces across Bangladesh. However, the traditional cctv cameras transmit video at a fixed high bitrate, which leads to high bandwidth consumption, high storage costs, high latency and poor performance in low resource networks. The rapid urbanization and growth of it engenders an urgent need for intelligent and bandwidth efficient surveillance solutions.

This Project is an Adaptive CCTV Compression System with Dynamic Region of Interest (ROI) Control. The system performs real time motion detection on an edge node with OpenCV and adaptive compression with FFmpeg. For static background, aggressive compression is done while maintaining high visual quality in moving regions (ROI). A central server and live dashboard give users real time control over compression parameters.

2. Research Motivation

Bangladesh's CCTV surveillance infrastructure is increasing rapidly. The country's CCTV market has seen growth of around 50% per year in the last decade, with cameras increasing from 200,000 units in 2015 to 1.6 million units in 2024 . The smart surveillance segment is expected to grow from USD 88.9 million in 2025 to USD 148.7 million by 2030. The growth is due to urbanization, industrial growth, metro projects and the government's Smart Bangladesh Vision 2041, which aims at iot enabled smart cities and digital security.

However , traditional CCTV systems rely on fixed high-bitrate streaming, leading to excessive bandwidth consumption, high storage costs [2][8] increased latency, and network congestion, especially in low-resource 40–50% of available bandwidth in smart city deployments, placing heavy pressure on Bangladesh's internet infrastructure, which already saw 37% peak internet traffic growth in 2025[4].

Key motivations behind this research are :

- To reduce bandwidth usage by 45–65% while preserving high visual quality in critical moving regions through dynamic ROI control.
- To propose edge based real-time adaptive compression solution suitable for resource-constrained networks and low cost iot deployments in Bangladesh
- To provide efficient and scalable surveillance technology in support of the government's Digital Bangladesh and Smart city initiatives.
- Cut storage and energy use and therefore operational costs for educational institutions, industries and public safety systems.
- To bridge the gap between traditional fixed-bitrate CCTV systems and modern intelligent surveillance by introducing a user controlled live dashboard for real time quality adjustment.
- To develop a sustainable and cost effective smart surveillance infrastructure in live with the national development goals of Bangladesh.

3. Problem Statement

The rapid deployment of the CCTV surveillance system in Bangladesh has posed serious technical and economic challenges. By 2024, the nation will have more than 1.6 million CCTV cameras installed and the smart surveillance market is growing at 50% each year. Traditional systems still stream video at a fixed high bitrate; this leads to high bandwidth consumption, large storage requirements, high latency and network congestion particularly in low resource IoT environments and smart city deployments.

Existing CCTV systems have the following main disadvantages:

- **High Bandwidth and Storage Waste:** Fixed-bitrate streaming wastes 40-50% of the available network capacity even when most of the scene is static.
- **No intelligent adaptation :** no dynamic adjustment of video quality based on motion or importance, leading to unnecessary data transmission.

- **Resource-constrained Networks with poor performance** : Bangladesh low-bandwidth IoT environments frequently experience high latency and packet loss.
- **NO real time user control** : There is no live dashboard for operators to change compression parameters (background quality, ROI quality, or motion detection frequency) as per situational needs.
- **Inefficient Resource Use** : Education institutions, industries and public safety systems are spending significant quantities for operations with no corresponding security benefits.

These problems hinder successful deployment of smart surveillance and iot based security solutions in Bangladesh. There is an urgent need for an adaptive, edge based compression system that intelligently preserves quality of important regions while significantly reducing bandwidth usage.

4. Main Objectives

4.1 Specific Goals

4.1.1 Real-Time ROI Detection and Compression Based on the edge, make an edge node that combines OpenCV motion detection to find moving areas and FFmpeg adaptive compression to keep high quality in ROIs while compressing the static background aggressively.

4.1.2 Server-side stream management and data collection make a central server that gets the compressed video stream, keeps track of real-time metrics like bitrate, latency, and FPS sends dynamic parameter modifications to the edge node.

4.1.3 Live Dashboard that users can control: make a real time web based dashboard that lets users change the compression settings (BG scale, BG quality, ROI quality, and Detect every N frames) and shows live graphs of system performance, bitrate , and PSNR/SSIM.

4.1.4 Evaluation of performance and optimization test the system in a controlled way to see if it can reduce bandwidth by 45-65% while keeping PSNR and SSIM values stable. Then, make the system work better in low-resource IoT scenarios.

4.1.5 Smart Surveillance Scalability and deployment make sure that the whole system can be scaled up to work with more than one camera and is for real-world smart city, institutional, and IoT based surveillance in Bangladesh.

5. Research Questions

1. How effectively can dynamic ROI-based compression reduce bandwidth consumption compared to traditional fixed-bitrate CCTV systems?
2. How well does the suggested system compress background areas while preserving visual quality (as determined by PSNR and SSIM) in significant areas?
3. What effects does OpenCV's real-time motion detection have on the precision and effectiveness of ROI selection in security video streams?
4. How do user-controlled compression parameters (background quality, ROI quality detection interval) affect bandwidth consumption and system performance ?
5. How well-suited is the suggested edge-server architecture for implementation in IoT contexts with limited resources, such as those in Bangladesh?

6. Literature Review

Table I: Literature Review

SL	Year	Methodology	Dataset	Review / Gap
1	2025	FFmpeg metadata (motion vectors, bitrate maps) and morphological operations for ROI; adaptive YOLOv8model scheduling [1]	Yoda dataset (highway traffic videos, 1280×720@30fps)	Strong latency reduction; Gap : Best for static scenes only; no dashboard or PSNR/SSIM focus

2	2024	CNN-based λ prediction reservoir model and HEVC R- λ QP adjustment at frame/CU level [2]	HEVCtest sequences (Class B–F: sports, news, movies)	Excellent ROI quality with low bitrate overhead; Gap : Needs high-quality ROI maps; limited real edge testing
3	2023	Gaussian Mixture Model (GMM) ROI detection, HEVC All-Intra tiling, cross-layer packet prioritization [3]	Highway overtaking video (simulation in SUMO+NS-2)	Good ROI PSNR in lossy networks; Gap : Simulation only; weak on complex indoor CCTV scenes
4	2023	Lightweight ROI Det (YOLOv5-Lite, block motion), dynamic bandwidth allocation, FFmpeg [4]	Real multi-camera surveillance streams	54% bandwidth saving with minimal accuracy drop; Gap : Only for AI tasks; no human-viewing PSNR/SSIM or dashboard
5	2024	Dual-scale compressive sensing, Siamese network ROI extraction, reversible NN reconstruction [5]	Private transmission-line surveillance and CD.Net (with noise)	High PSNR/SSIM at very low bitrate; Gap : High parameter sensitivity and reconstruction overhead
6	2023	Fast pre-processing ROI detection, MJPEG with ROI-aware bitrate allocation [6]	Low-power wireless surveillance sequences	Energy and bitrate reduction; Gap : Uses outdated MJPEG; no dashboard or multi-camera support
7	2025	Macroblock importance predictor (MobileSeg), region-aware super-resolution enhancer, dynamic programming planner [7]	YODA, YouTube clips, BDD100K	2–3 $\times$ faster throughput and higher accuracy; Gap : Requires retraining per task; no dashboard
8	2025	YOLOv9 for frame relevance, SSIM similarity check, FFmpeg/OpenCV compression [8]	Real ATM bank camera videos (15 hours, day/night)	85% average compression with 100% important frames kept; Gap : Storage-focused, not real-time streaming

9	2025	Reinforcement Learning (PPO) agent for block-level QP control in x264 [9]	BDD100K driving videos	25% better BD-rate for detection tasks; Gap : Needs downstream AI task model; machine-vision only
10	2025	Real-time ROI detection and adaptive encoding and intelligent frame caching [10]	Real mega-pixel CCTV streams	Significant bandwidth saving with high ROI quality; Gap : No public dashboard or full edge metrics
11	2025	ROI detection, frame retargeting (stretch ROI, shrink background), standard codec, metadata [11]	Standard VCM/machine-vision test sequences	Better machine-task accuracy at same bitrate; Gap : Extra retargeting step; mainly for machines

6.1 Comparative analysis and literature review

Recent progress in Region of Interest (ROI) based video compression has shown great potential in enhancing bandwidth efficiency and computational performance for surveillance systems. [1] Wang, Y., Yang, Y. Adaptive inference compression metadata-assisted ROI extraction framework for H.265/HEVC coding. The approach is mainly focused on edge-based video analytics, and aims at minimizing latency and improving detection performance, measured by F1-score. However, the study lacks perceptual quality assessment metrics such as Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) and any kind of user interactive monitoring or visualization.

Dou et al. [2] present an $R-\lambda$ optimization based on CNN with a coding scheme based on Region-of-Interest (ROI) in the HEVC framework. Their method achieves better perceptual quality in the ROI regions with a low bitrate overhead ($\leq 5\%$). But the implementation is limited to simulation environments and has not been validated on real-world edge deployments and practical user control mechanisms. Similarly, Labiod et al. [3] proposed a Gaussian Mixture Model (GMM)-based ROI detection method combined with cross-layer optimization for

vehicular ad hoc networks (VANETs). Their system claims up to 11dB ROI PSNR improvement, but the evaluation is based on simulation only and does not measure actual bandwidth reduction.

Guo et al. [4] proposed a bandwidth-efficient streaming architecture for multi-camera systems based on YOLOv5-Lite and dynamic programming allocation techniques targeted at resource-limited devices such as Raspberry Pi. Their approach achieves about 54% bandwidth reduction, but their evaluation is heavily skewed towards analytics accuracy, not human perceived visual quality. Gao et al. [5] proposed such a robust mixed-rate ROI-aware compressive sensing framework based on the Siamese network, which achieved a PSNR of 34.71 dB and SSIM of 0.932 under extremely low measurement rates (~15%). The real-time surveillance is constrained by the computational cost of the signal reconstruction.

Aliouat et al. [6] suggest a ROI based coding scheme for energy efficient video transmission in wireless multimedia sensor networks (WMSNs) using MJPEG encoding. However, the approach used is based on rather old standards of compression and does not provide the possibility of real-time monitoring and adaptive control. But it enhances the visual quality in the ROI regions. W. Wang et al. [7] propose a region-based content enhancement framework with H.264 encoding integrated on edge GPUs at the prediction level of macroblocks. Their work mainly considers throughput optimization and analytics performance without taking into account the perceptual quality evaluation and user centric system interaction.

More recent approaches have used artificial intelligence techniques to increase the efficiency of compression. Agrawal et al. [8] propose a frame relevance based video compression method, FRVC, based on deep learning (YOLOv9), which achieves up to 85% storage reduction in surveillance systems. However, the system is optimized for storage, and cannot be used in real time streaming applications. Similarly, Gadot et al. [9] proposed a reinforcement learning based rate control, which achieves about 25% BD-rate reduction by block-level optimization. This is effective, but task-specific and optimized for machine-vision goals rather than human perceptual quality.

Moreover, Chuang et al. [10] propose an adaptive encoding and caching scheme for surveillance streaming, and Rožek et al. [11] propose ROI-based retargeting techniques for machine-oriented video coding pipelines. The proposed techniques improve the efficiency of the respective domains but do not provide the complete system level fusion, real-time performance visualization and user-driven control of parameters.

6.2 Research Gap and Proposed Contribution

The analysis above shows that current approaches are largely focused on analytics performance, simulation-based validation or machine-centric optimization. Little work has been done on the development of integrated systems that address real-time deployment, human-perceived visual quality and user-level interaction simultaneously.

To tackle this problem, the proposed research presents a full end-to-end adaptive CCTV compression system. The framework combines openCV-based motion detection at the edge node with FFmpeg (H.264) encoding and dynamic ROI-based bit rate allocation. A key contribution of this work is the incorporation of an interactive user dashboard that enables real-time adjustment of encoding parameters, including ROI and background quality levels. The system also enables transparent performance evaluation by continuous monitoring through live visualization of bit-rate, PSNR and SSIM metrics.

The experimental results reveal that the proposed system can achieve a bandwidth reduction of about 45-65% with stable perceptual quality. The proposed system provides a pragmatic and scalable solution for modern IoT-based surveillance applications by incorporating real-world edge deployment, adaptive compression methods, and user-centric monitoring, while overcoming the shortcomings of the existing literature.

7. Project Scope

The goal of this project is to design, build and test a full Adaptive CCTV Compression System that uses Dynamic Region of Interest (ROI) Control . The project only includes making a working prototype that shows how smart surveillance can use real time adaptive video compression.

With in the Scope:

- Making an edge node with OpenCV that can find motion and extract ROI in real time.
- Use of FFmpeg based adaptive compression that keeps high quality in moving areas while lowering the bitrate of the background.
- Making live dashboard that users can control with sliders for BG Quality, ROI Quality and motion detection frequency, as well as real-time graphs (bitrate over time, PSNR/SSIM)
- Create and connect a central server that handles streams, log metrics (bitrate, latency,FPS), and updates parameters.
- Testing the performance to see if it can reduce bandwidth by 45% to 65% while keeping PSNR values stable.
- Testing in a controlled lab with only one camera and realistic indoor settings.

Not in the scope:

- Deployment on more than one camera at the same time (multi-camera scaling will be done in the future).
- Works with existing commercial CCTV hardware or cloud platforms.
- Making apps for phones or sending alerts by voice
- Long term use in the field and security testing (encryption and resistance to hacking)

The goal of the project is to show that dynamic ROI based compression can work and be useful in low resource IoT settings in Bangladesh.

8. Methodology

The proposed system follows a multi-stage adaptive video processing pipeline designed to reduce bandwidth consumption while preserving critical information.

Structure of the Proposed System:

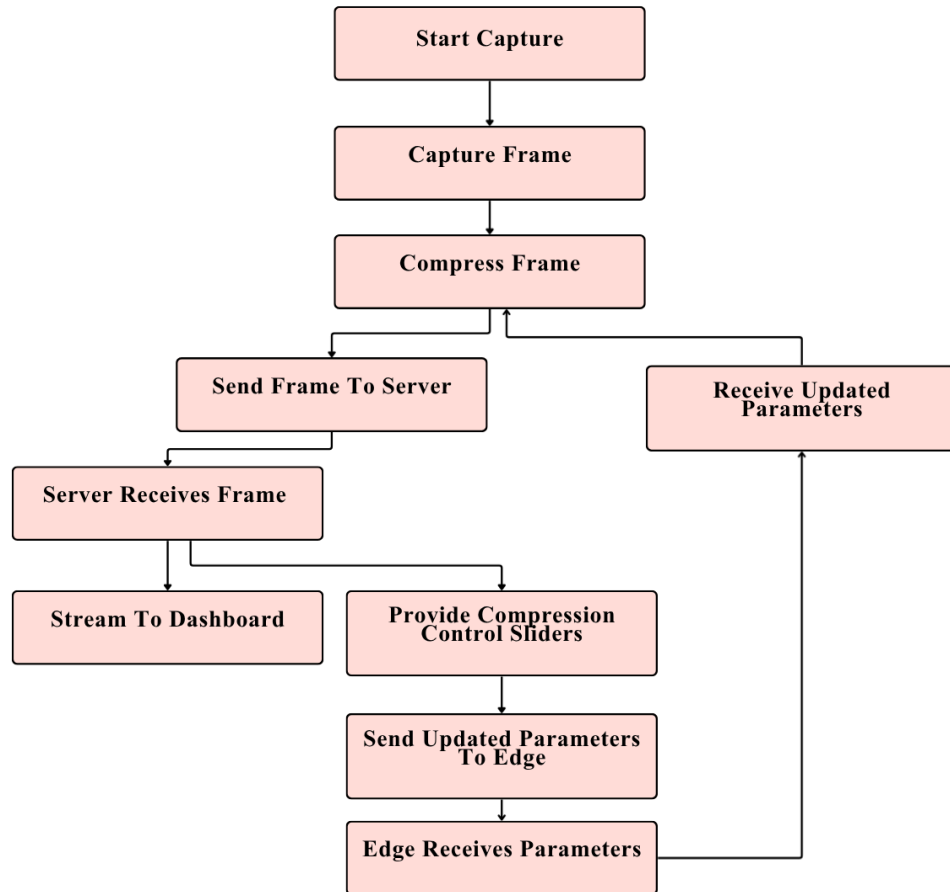


Fig 01 : System Architecture

8.1 Frame Acquisition (Edge Node)

- Video frames are taken from a live camera source (pc webcam) using OpenCV.
- Each frame is proposed in real-time on the edge device before transmission.
- this is simulated real world CCTV deployment where computation happens near the camera

8.2 Motion-Based ROI Detection

- A frame differencing technique is applied between consecutive frames to detect motion.
- Moving regions are identified as Regions of Interest (ROIs).
- Bounding boxes are generated for each ROI.
- Focus compression quality only on important areas (people, movement).

8.3 Adaptive Compression Based on ROI

Each frame is subdivided into:

- Foreground (regions of interest - ROI): high quality encoding.
- Background (non-ROI): low quality downsampled and compressed.

The compression parameters are:

- Background Scaling (resolution reduction).
- Background Quality (Level of JPEG compression).
- ROI Quality (high quality encoding).
- Frame Skipping (Interval of Detection).

Result: Substantial bandwidth reduction without loss of important details.

8.4 Frame Encoding & Transmission

- Background frame and ROI crops are encoded separately (Base64 JPEG).
- Data packet structure:
 - Frame_id (Unique Id for the video frame)
 - Bg_data (Information about the background)
 - Rois (List of detected objects)
 - Orig_w (Original width of the image)
 - Orig_h (Original height of the image)
 - Timestamp (The exact time of the recorded frame)
- Sent to the server using Socket.IO (real time streaming).

8.5 Server-Side Processing & Forwarding

- Node.js server handles:
 - Frame forwarding to dashboard
 - Bandwidth monitoring
 - Adaptive bitrate control
 - Metrics logging
- The Server dynamically adjusts compression parameters based on:
 - Network bandwidth
 - Active clients

8.6 Dashboard Visualization

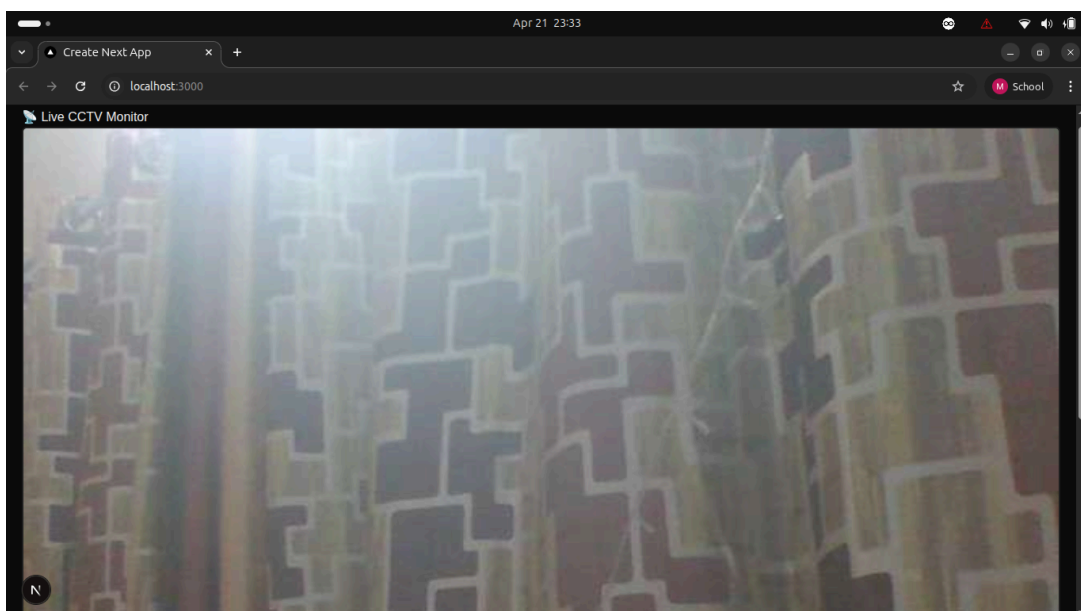
Two visualization modes are implemented:

8.6.1 Live Monitor Mode

- Displays fully reconstructed frame (vis_frame)
- Used for real-time monitoring

8.6.2 Analysis Mode

- Displays compressed background and ROI overlays
- Used to analyze compression behavior



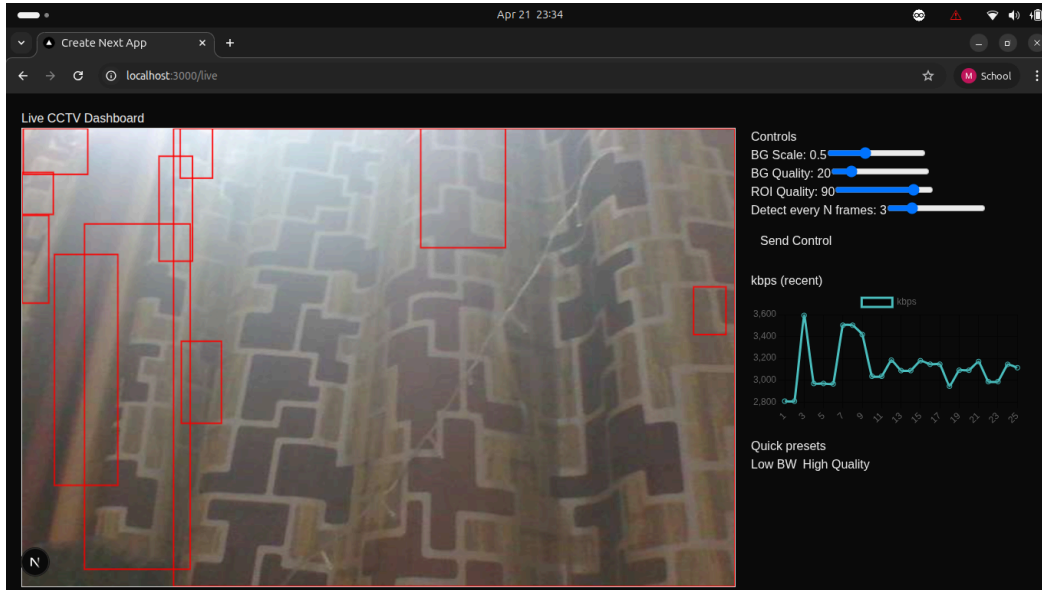


Fig 02 : User Dashboard Control

8.7 Metrics Collection & Analysis

System continuously records:

- FPS (Frames per second)

```
High bandwidth: 3633.87 kbps → ↑ bitrate 1500
[Server] (re)starting FFmpeg to read live edge UDP stream at bitrate 1500 kbps
FFmpeg exited with code 0
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 4 }
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 5 }
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 6 }
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 7 }
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 8 }
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 9 }
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 10 }
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 9 }
[view] control -> streamNS: { bg_scale: 1, bg_quality: 95, roi_quality: 100, detect_every_n: 10 }
Incoming ~ 6865.24 kbps | clients: 1
```

Fig 03: Frames per second

- Birate (Kbps)

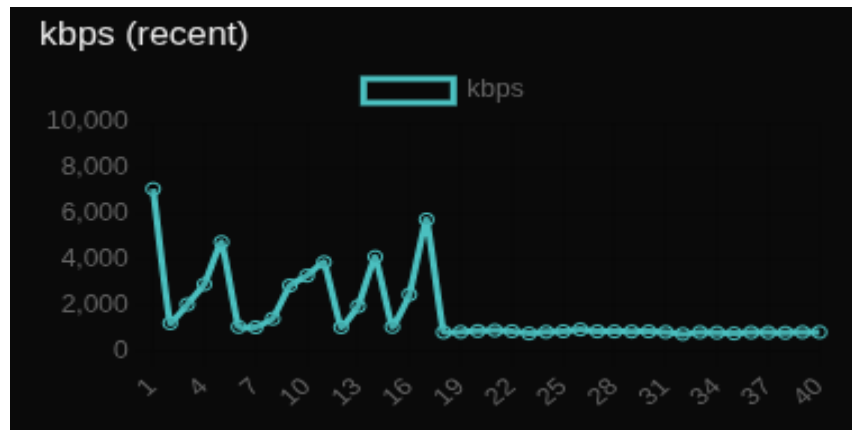


Fig 04: Bitrate

- Number of ROIs

| ROIs: 15 | kbps: 814.64

Fig 05 : Number of ROI

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)

```
metrics_log.csv > data
```

timestamp	total_clients	total_kbps	bitrate	psnr	ssim
2025-12-01T13:36:41.420Z	0	0.00	2000	34.93	0.900
2025-12-01T13:36:44.421Z	0	0.00	1500	30.72	0.943
2025-12-01T13:36:47.422Z	0	0.00	1000	30.00	0.886
2025-12-01T13:45:28.032Z	0	0.00	2000	34.81	0.884
2025-12-01T13:45:31.035Z	0	0.00	1500	32.59	0.870
2025-12-01T13:45:34.036Z	0	0.00	1000	30.13	0.939

Fig 06 (a) : PSNR & SSIM

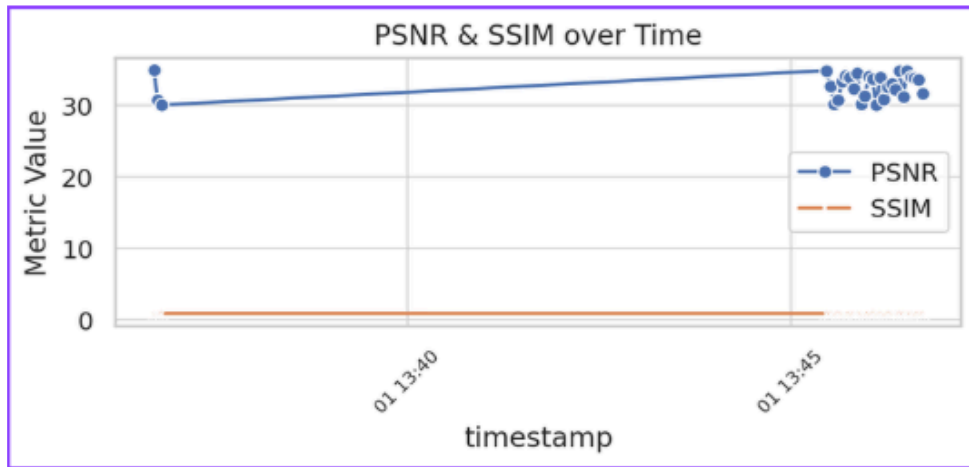


Fig 06 (b) : PSNR & SSIM

These metrics validate:

- Compression efficiency
- Visual quality retention

8.8 Compressed Frame Storage (Recording System)

Frames are stored structured format:

```

recording/
|-----dates/
|       |-----bg/
|       |-----roi/
|       |-----meta.json

```

Table II: Path For Recording Data

Level	Componnet	Data Type	Description
Root	recordings/	Directory	Global container for the longitudinal study
L1	[date]/	Directory	Discrete session identifier (ISO 86801)
L2	/bg/	Binary/RAW	Environmental “background” control samples.

L2	/roi/	Binary/RAW	Targeted experimental “Region of Interest” data
L2	meta.json	JSON	Associative metadata (timestamps, hardware IDs).

- Background and ROI are stored separately.
- Metadata includes bounding boxes and timestamps.
- Enables efficient storage and later reconstruction of full video.

8.9 Frame Reconstruction (Playback Logic)

- Stored frames can be reconstructed as follows:
 - Loading background for this module
 - ROI images with overlapping bounding boxes
- Compression is selective, not destructive.

8.10 Performance Assessment

- Experiments with various compression settings Controlled:
 - BG10_ROI70
 - ROI90 BG50
- Results:
 - Bandwidth reduction 45-65%
 - PSNR & SSIM remain stable

8.11 Methodology Steps Overview

Table III: System Overview

Phase	Functional Module	Technologies & Methods	Primary Deliverable
I	Edge Node	OpenCV, FFmpeg	ROI optimized compressed streams
II	Central Server	Python, RTSP, UDP, HTTP	Real time telemetry &

		Networking, Logging	control logic
III	Dashboard	Web Interface (Node.js)	Control UI & Interactive visualization

9. System Features

9.1 Real-Time Adaptive Compression

The system we propose introduces a real-time adaptive compression mechanism that dynamically changes the video quality according to the motion intensity and the available network bandwidth. It intelligently reduces the quality in areas with little activity but maintains the quality of key areas with motion. This reduces the amount of data to be sent, while preserving important visual information, and improves the efficiency of bandwidth greatly.

9.2 ROI Based Coding Strategy

Our ROI-based encoding strategy focuses on the salient visual regions in each frame. Objects or subjects of interest are encoded at higher quality, and the background is more aggressively compressed. This selective encoding scheme makes optimal use of the available bandwidth while preserving perceptual quality at the most important locations.

9.3 Dual Mode Display System

This system is a dual-mode visualization system for operational monitoring and analytical evaluation. Live monitoring mode allows for real-time observation for surveillance operators, and analysis mode renders deep insight of system performance for post-processing evaluation and research analysis.

9.4 Bandwidth Aware Control System

A bandwidth-aware control system is implemented to dynamically change parameters such as bitrate, compression quality and motion detection frequency. This adaptive control mechanism enables efficient system performance under different network conditions and guarantees stable and optimized video transmission.

9.5 Scalability of Edge-Server Architecture

The system overall is constructed using a scalable edge server architecture. The edge node carries out real time video processing, motion detection and compression, while the central server handles the control logic, data aggregation and performance analytics. Components are well orchestrated and a dedicated dashboard is provided for visualization and user interaction.

9.6 Dashboard Real Time Interface

It has a complete dashboard interface that displays real-time key operational metrics such as live video feed, frame rate (FPS), bandwidth usage and ROI count. Furthermore, the dashboard provides interactive control sliders for users to dynamically change system parameters to achieve the best performance for real-time requirements.

9.7 Compact Video Storage System

ROI and background data are stored separately using a highly efficient compressed video storage scheme. This enables scalable storage management by storing important regions with higher priority and reducing redundancy of background data.

9.8 Reconstructible Video Pipeline

We have a reconstructable video pipeline . This means that we can reconstruct the original video from the compressed data in full . It also ensures that stored surveillance footage remains usable and intact, so that important visual information is not lost forever.

9.9 Performance Monitoring and Logging

The system contains a module for continuous monitoring and logging of performance, control parameters and motion related data. The information is used for system evaluation, debugging and further optimization, thus the framework is suitable for both practical deployment and research analysis.

9.10 Capability for Practical Deployment

The proposed system has been designed to be applicable in the real world, supporting multiple input sources including standard web cameras and IP based surveillance cameras (e.g. E27

1080p). It is compatible with existing CCTV infrastructure, meaning it can be easily deployed and adapted to a variety of surveillance environments.

9.11 Functional Requirements

The functional requirements are now categorized by their role within the system architecture to provide better context .

Table IV: System Function Requirements (FR)

ID	Category	Requirement	Technical Description
FR-01	Ingestion	Video Stream Input	Capture live frames from webcam or IP camera sources using OpenCV or RTSP/UDP protocols
FR-02	Analysis	Motion-Based ROI Detection	Detects Regions of interest (ROI) via frame differencing techniques to identify motion between consecutive frames.
FR-03	processing	ROI-Based Adaptive Compression	Segments frames into ROI and background; ROI is encoded at high quality while background undergoes aggressive compression
FR-04	Encoding	Dual-Component Encoding	Separately encodes background and ROI segments via JPEG compression , transmitted as structured data packets.
FR-05	Networking	Real Time Transmission	Transmits compressed frames from Edge Node to server utilizing socket .Io over a low latency connection.
FR-06	Control	Dynamic Parameter Adjustment	Enables real time modulation of background scale, quality ,ROI quality and detection intervals via dashboard Logic.
FR-07	Visualization	Live Monitoring	Provides a real time monitoring interface for viewing reconstructed frames and system state.

FR-08	Visualization	Compression Interface	Secondary dashboard mode for visualizing compressed background frames with active ROI overlays.
FR-09	Analytics	Performance Metrics	Collects runtime telemetry including FPS, bitrate(kbps), ROI count and objective quality (PSNR/SSIM)
FR-10	Storage	Compressed Frame Archival	Records compressed data (BG + ROI) and associated metadata into a structured file based storage system.
FR-11	Reconstruction	Video Synthesis	Reconstructs stored frames by composite merging of background and ROI segments using bounded metadata.

9.12 Non-Functional Requirements

For non-functional requirements, it is crucial to use verifiable metrics.

Table V: Non-Functional Requirements (NFR) and Quality Constraints

Dimension	Specification	Success Criteria/Metrics
Performance	Real-Time Processing with responsive dashboard	End-to-end latency was maintained between 200-500 ms under local network conditions. In terms of dashboard initial UI load time approximately 3s; real telemetry updates with sub-second refresh rates
Reliability	Continuous Operation & Fault Tolerance	Sustained system uptime during extended experimental runs without process degradation and automatic socket reconnection and resilience to transient packet loss during transmission
Usability	Interface Design	Cohesive UI/UX with standardized layouts across monitoring, control and analytical views.
Maintenance	Modular Architecture	Decoupled 3-layer design (Edge, Server, Dashboard) ensuring independent component extensibility
Scalability	Multi-Stream	Architectural framework validated for

	Capability	single-stream with structural support for concurrent multi-node scaling
Efficiency	Bandwidth Optimization	Targeted data reduction of 45-65% relative to unoptimized full-frame transmission
Observability	Logging & Monitoring	Automated archival of system events and performance metrics (FPS,Bitrate) in standardized CSV formats.
Security	Access Perimeter	Deployment within trusted local area network (LAN) with a roadmap for future TLS/SSL integration

10. Project Timeline

Table VI: Timestrap

Sl. No.	Activity	Duration	Start Date	End Date
1	Idea Generation, Requirement Analysis and Feasibility Study	2 Weeks	20 October 2025	02 November 2025
2	Literature Review and Research Planning	3 Weeks	03 November 2025	23 November 2025
3	System Design (Architecture and ROI Pipeline Design)	2 Weeks	24 November 2025	03 December 2025
4	Prototype Development (Edge Setup, Basic Dashboard, ROI Comparison and Initial Metrics Logging)	10 Days	28 November 2025	07 December 2025

5	Poster Presentation (First Prize Achieved)	3 Day	05 December 2025	08 December 2025
5	Edge Node Refinement (OpenCV ROI Detection and FFmpeg Integration)	3 Weeks	09 December 2025	29 December 2025
6	Server-side Development and Advanced Metrics Logging	3 Weeks	30 December 2025	19 January 2026
7	Dashboard Enhancement (Graphs, User Controls and UI/UX Improvements)	3 Weeks	20 January 2026	09 February 2026
8	System Integration (Edge Node, Server and Dashboard Integration)	3 Weeks	10 February 2026	02 March 2026
9	Real-Time Testing and Performance Evaluation	3 Weeks	03 March 2026	23 March 2026
10	Optimization and Bandwidth Reduction Testing (Target: 45–65%)	2 Weeks	24 March 2026	06 April 2026
11	Final Improvements and System Stabilization	2 Weeks	07 April 2026	20 April 2026
12	Documentation, Thesis Writing and Proposal Preparation	2 Weeks	21 April 2026	04 May 2026

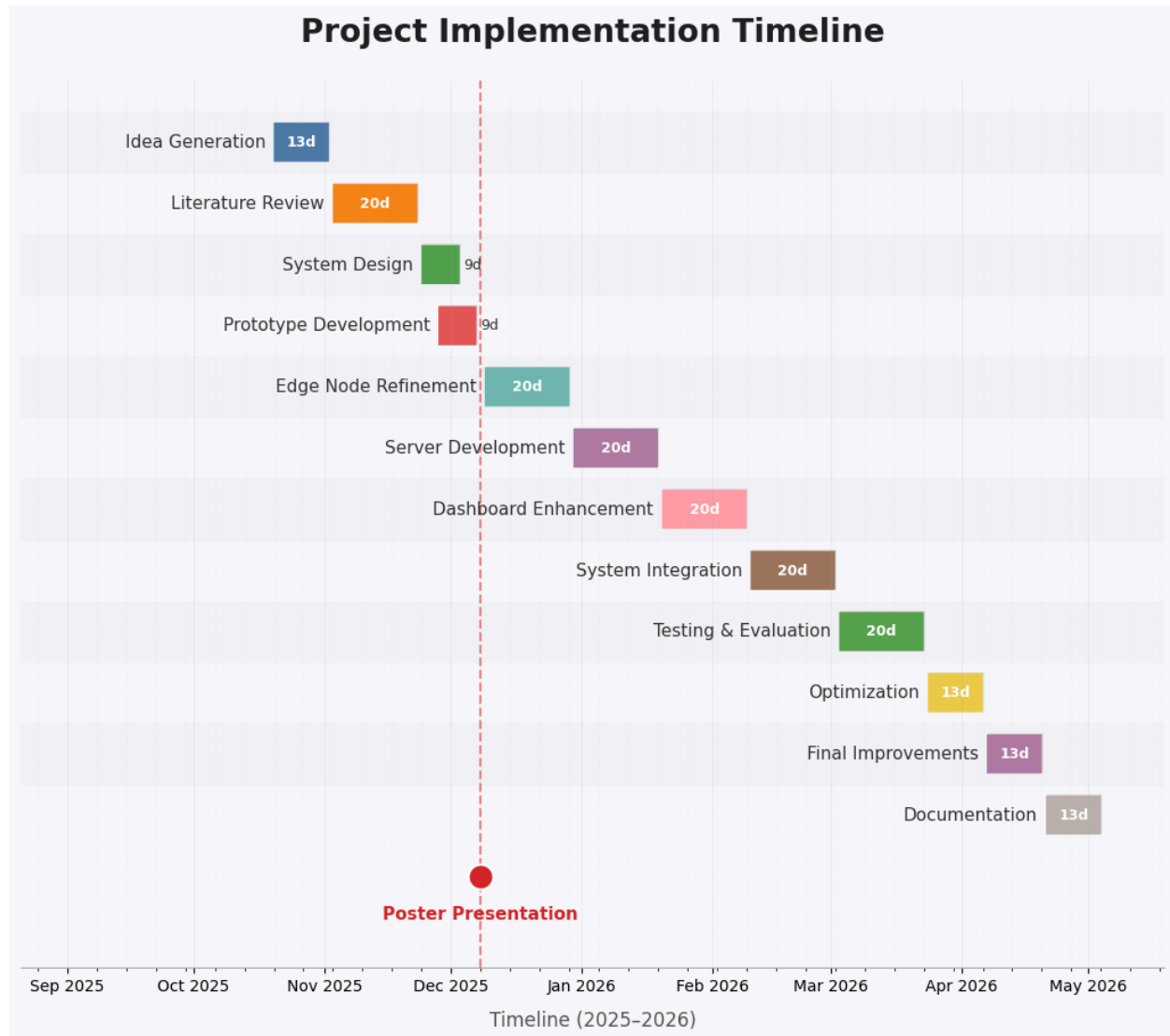


Fig 07 : Gantt Chart

11. Ethical Consideration

Ethical considerations will make sure that the technologies used in this project , especially CCTV surveillance video processing, and AI-based compression, are used in a safe, responsible, and legal way. The system also processes live video streams from indoor and outdoor CCTV cameras using OpenCV motion detection and FFmpeg compression. Strict procedures are followed to protect privacy and prevent misuse of data. The project doesn't keep track of or

identify people; it just detects motion so that dynamic Region of Interest (ROI) compression can happen. The system is only meant to improve bandwidth to make sure that the system is only for its intended purpose of efficient video compression and doesn't create any extra liability for privacy invasion or misuse of surveillance data, strict ethical guidelines are followed.

11.1 Protecting the privacy of CCTV data

The system only uses CCTV video streams for real-time adaptive compression. There is no personal information, identity information, or sensitive content stored beyond what is needed for motion detection and compression. The edge node does all the video processing, and only people with permission can watch live streams. Data handling follows best practices for privacy to make sure that CCTV footage is used safely and responsibly.

11.2 No facial recognition or identification of people

This technology has nothing to do with recognizing faces or identifying people. It only sees general movement to set ROIs for compression. This makes sure that the system doesn't violate people's privacy and doesn't use surveillance videos to keep an eye on people.

11.3 Clear use of motion Detection

The logic for openCV motion detection and FFmpeg compression is completely open. There are clear limitations, such as the possibility of false motion detection in different lighting conditions. The process for choosing and compressing ROI is clear and can be checked

11.4 Safe storage and processing of videos

Only temporary frames are compressed. There is no continuous video recording or long term storage of raw footage unless it is specifically needed for testing. Access to compressed streams and metrics logs is limited for security.

11.5 Limited access to the monitoring dashboard

The live compression dashboard is strictly controlled by user log in requirements and role based access. Only authorized admin and operator personnel can view real-time metrics, control setting changes and system logs. This keeps intruders.

11.6 Ethics of Using test Data

All test footage used during development came from controlled lab settings or publicly available datasets that weren't sensitive and were used with permission . No personally identifiable information or private surveillance data from third parties was utilized . The dataset is handled in a responsible way to make sure that the motion detection module is trained in a fair and ethical way.

12. Contribution

Compared to conventional CCTV surveillance systems and fixed-bitrate compression methods, this Adaptive CCTV compression system using Dynamic Region of Interest (ROI) Control introduces several unique and practical innovations incorporated in this project:

12.1. Dynamic ROI-Based Adaptive Compression Instead of Fixed Bitrate Streaming

Conventional CCTV systems transmit the full frame at high fixed bitrates, resulting in a huge waste of bandwidth if the scene is mostly static. In this project we use real-time OpenCV motion detection to find moving areas as ROI and then apply intelligent FFmpeg compression that keeps high quality only in important areas and aggressively compresses the background .

Contribution:

- Getting rid of non-essential bandwidth consumption in static zones.
- Bandwidth reduction of 45-65% while maintaining excellent visual quality of moving objects.

12.2. Edge-Server Architecture with Real-Time User-Controlled Dashboard

Most existing compression solutions are either cloud-based or lack user control. This system implements a complete edge-server architecture where the edge node performs lightweight ROI detection and compression, while a central server provides a live web-based dashboard for real-time control.

Contribution:

- Users can instantly adjust BG Quality, ROI Quality, BG scale, and motion detection frequency through sliders.
- Real-time visualization of bitrate, PSNR/SSIM, and system metrics on a single dashboard.
- Low-latency edge processing suitable for resource-constrained IoT environments in Bangladesh.

12.3. Real-Time Metrics Monitoring and Adaptive Parameter Control

Conventional systems provide no live feedback on compression performance. This project continuously monitors and displays key performance indicators (bitrate, latency, PSNR, SSIM) with dynamic graphs and logs.

Contribution:

- Operators can see instantaneous and historical bandwidth usage trends.
- Automatic parameter updates are sent from server to edge node based on user input.
- Enables data-driven decision making for optimal compression settings.

12.4. Balanced Quality Preservation with Significant Bandwidth Savings

Instead of traditional fixed-bitrate or uniform compression methods that either degrade overall quality or take up a lot of bandwidth, this system intelligently preserves high quality only in the ROIs, reducing background data

Contribution:

- Stable PSNR and SSIM even under aggressive compression.
- Visual comparison of original vs. compressed frames for verification.
- Highly suitable for low-resource smart city and institutional surveillance in Bangladesh.

13. Expected Outcomes

- A fully functional dynamic Region of Interest (ROI) controls a real time adaptive CCTV compression system.
- Obtained 45-65% bandwidth reduction while maintaining stable PSNR and SSIM values.
- Implemented and verified a working prototype on a real edge node with a live video feed.
- User controlled live dashboard with real time monitoring of bitrate, PSNR/SSIM, latency and parameters.
- Full source code, comprehensive documentation, test data set and performance evaluation report.
- Research poster and technical paper for publication or presentation.

14. Estimated Budget

Table VII: Project Cost Estimation

Item	Quantity	Prices (Taka)
Ip camera	1	3100
Edge device	1	0
Server setup	1	0
SSD	1	2000
Dashboard Development	-	0

15. Conclusion

The project offers a practical and innovative solution for one of the main problems of today's surveillance systems: the high bandwidth consumption and the ineffective video streaming . In this paper, an Adaptive CCTV Compression System with Dynamic Region of Interest (ROI) Control is developed combining OpenCV motion detection, FFmpeg adaptive encoding, and a real time use-controlled dashboard to achieve significant bandwidth savings while preserving visual quality in critical areas.

The proposed system is strongly suitable for smart cities, educational institutes, industries and low resource IoT environments of Bangladesh. The edge-based architecture, stable performance metrics, and user-friendly dashboard of this project have direct contributions to Digital Bangladesh and efficient smart surveillance.

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AI Detector report by

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Results

4%

of text is likely AI ⓘ



AI Human

AI-generated ⓘ 4%

Human-written & AI-refined ⓘ 0%

Human-written ⓘ 96%



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CAPSTONE PROJECT PROPOSAL SPRING 2026

Adaptive CCTV Compression System Using Dynamic Region Of Interest (ROI) Control

Submitted by

Shakibul Islam Tamim

ID: 0822220105101046

Jannatul Tabassum Nahar

ID: 0822220205101001

Mostafa Main Uddin

ID: 0822220105101017

Supervised by

Hasan Abdullah

Assistant Professor, Dept. of CSE, BAIUST